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Experimental study and thermodynamic assessment of the dysprosium-hydrogen binary system

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ABSTRACT

The dysprosium-hydrogen (Dy-H) binary system in the full range of H/Dy = 0 to 3.0 is investigated by accurate pressure composition isotherm (PCI) measurement, including a PCI measurement of the β-DyH_2/γ-DyH_3 transition for the first time. The Dy/β-DyH_2 biphasic region shows well defined plateaus in the range of H/Dy = 0.41 to 1.65 at 650 °C and gradually shrinks to 0.54 to 1.23 at 900 °C, with transition enthalpy of 213.1 ± 2.8 kJ/mol H_2. The β-DyH_2/γ-DyH_3 biphasic region also shows well defined plateaus from 250 °C to 325 °C, but with much narrower composition region of H/Dy from 2.3 to 2.9. The transition enthalpy is 79.3 ± 2.2 kJ/mol H_2. The phase diagram and thermodynamic parameters are calculated with the CALPHAD method using the measured PCI data. The obtained results are in very good agreement with our experimental data and the published data reported in literature. The results provide more comprehensive and quantitative insights into the key thermodynamic properties of the Dy-H system.
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1. Introduction

Rare earth–hydrogen (RE-H) systems have attracted attention in recent years both due to their exceptional fundamental properties and wide range of applications in fields such as hydrogen storage, permanent magnetic and switchable mirror [1–3]. To utilize the full technological potential of the RE metals, it is essential to have a fundamental knowledge about the RE-H systems, such as information about the phase equilibrium and the related thermodynamic data. The thermodynamic properties of the La-H [4], Ce-H [5], Nd-H [6], Sm-H [7],Gd-H [8] and Er-H [9] systems have been well studied by various experimental and computational techniques. However, we still are lacking the constitution of a thermodynamic database able to cover the needs for calculation of multi-component metal-hydrogen systems [10].

Dysprosium-Hydrogen (Dy-H) system is one of the most attractive RE-H systems nowadays used in a wide variety of applications. The coercivity of Nd-Fe-B magnets can be effectively enhanced by the addition of a small amount of Dy [11]. Dy hydride as Dy sources have been widely studied in these magnets due to the high cost and inconvenient preparation of pure Dy powders [12–14]. The H content of doped Dy hydride has significant effect on the magnetic properties and the resultant microstructures of Nd-Fe-B sintered magnet. In addition, Dy film can serve as smart coatings which can switch their optical properties reversibly between reflective and transparent states by hydrogenation and dehydrogenation [15]. It is also of high interest to understand the interactions between Dy and H in fields such as hydrogen storage alloys [16] and recycle use of electronic waste containing Dy metal [17].

However, the available Dy-H binary phase diagram from Mulford's work [18] is very rudimentary and just presenting the formation of two nonstoichiometric hydride phases β-DyH_2 and γ-DyH_3. More importantly, the phase boundaries are roughly evaluated from a generalized phase diagram for heavy RE-H systems in the absence of detailed experimental measurements.

For a Metal-Hydrogen system, the best approach to obtain thermodynamic data and deduce partial phase diagram is the accurate and detailed measurement of pressure composition isotherms (PCI) curves. The van't Hoff plot correlating the plateau pressures with temperature can be used to determine the enthalpy and entropy of the hydrogenation process. The phase

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transformation, or more specifically, the phase boundaries in the phase diagram are critically determined by the initial and terminal points of the pressure plateaus. Therefore, to obtain accurate experimental binary M-H phase diagram using the PCI method, the initial and terminal points of the plateaus have to be accurately determined. To the best of our knowledge, such high quality experimental data is absent currently.

In addition to the experimental approaches, the CALPHAD (CALculation of PHase Diagrams) method, using thermodynamic models coupled with key experimental measurements, is an efficient semi-empirical technique to get better knowledge of the phase diagrams and thermodynamic properties. Contrary to many purely metallic systems, only a limited number of RE-H systems have been optimized using the CALPHAD method [10,19]. Accurate PCI measurements can be a great help in this method to optimize the parameters describing the Gibbs energies of the different phases.

The purpose of the present work is to present the results of PCI equilibria performed for the Dy-H system with higher accuracy, particularly for the points determining the phase boundaries, to provide more quantitative understanding on the fundamental thermodynamics of this system. A first assessment of Dy-H system has been carried out by the CALPHAD method on the basis of our PCI measurements and critically collected experimental data from the literature.

2. Review of experimental data

Dysprosium has a hexagonal crystal structure at room temperature and transforms into a body centered cubic structure at 1381 °C [20]. Two nonstoichiometric hydride phases are stable in Dy-H system: DyH_2 with CaF_2 crystal structure (face centered cubic, with hydrogen occupying the tetrahedral sites) and DyH_3 with HoD_3 crystal structure (hexagonal closest packing Dy substructure with hydrogen occupying all the tetrahedral sites, part of the triangular sites displaced toward octahedral positions, and part of the triangular positions) [21–24].

As typical heavy rare earth-hydrogen systems, the PCI curve for the Dy-H system exhibits two pressure plateaus in the Dy+β-DyH_2 and β-DyH_2+γ-DyH_3 mixed phase regions. The PCI curves of Dy-H system have been measured by Beck between the Dy solid solution hcp(Dy) and the dihydride β-DyH_2 in the temperature range 750–1000 °C [25]. However, the number of the measurement points was not sufficient, thus the initial and terminal points of the plateaus corresponding to the transition from hcp(Dy) to β-DyH_2 were evaluated without sufficient accuracy. For example, the solubility of H in Dy can be only confirmed in the range of H/M = 0.36–0.52 at 750 °C and H/M = 0.47–0.64 at 900 °C. The metal-rich phase boundary of nonstoichiometric β-DyH_2 was determined in the range of H/M = 1.46–1.77 at 750 °C and H/M = 0.78–1.28 at 900 °C. These experimental data cannot be used to accurately infer the phase boundaries and should be rejected from the optimization in CALPHAD technique. As a confirmation, the Nd-H phase diagram calculated by Luo et al. [6] deviates from Mulford's PCI measurement [26] because the initial and terminal points of the plateaus were evaluated without sufficient accuracy. And these PCI data were not used for the optimization of the Nd-H system to ensure the accuracy of the calculated phase diagram. McQuillan has directly measured the van't Hoff plot in this two-phase region by ascending and descending temperature but without providing the exact hydrogen content [27]. Besides, no PCI measurements between the transition of the two hydrides (β-DyH_2 and γ-DyH_3) were found in the literature.

In addition, several X-ray diffraction and Neutron diffraction measurements have been made in the β-DyH_2+γ-DyH_3 two-phase region at room temperature to determine precisely the phase boundary [21,28]. The calorimetric measurements concerned with low-temperature specific heat of DyH_2-DyH_3 system and combined TG-DTA technique has been performed by Mintz [29].

3. Experimental

The specimens used in this study were pure distilled dysprosium ingots (purity of 99.95%). The gaseous impurities composition of Dy (wt.%) is 0.006 N, 0.015 O, 0.003 H.

PCI curves have been measured by a home designed stainless steel constant-volume Sieverts apparatus with sufficient sensitivity and accuracy. The whole system consists an UHV reactor with a base pressure of about 8×10^{-8} Torr (~1.1×10^{-5} Pa) and a low leaking rate of 8.4×10^{-4} Pa cm^3 s^{-1}. To accurately determine the low equilibrium hydrogen pressures in the hcp(Dy)+β-DyH_2 two phase range, two capacitance pressure gauges with ranges from 1.33×10^{-2} to 1.33×10^{2} Pa and 1.33×10^{1} to 1.33×10^{5} Pa respectively, were used to monitor the pressure in the system. A computer controlled data acquisition system has been used to record the hydrogenation kinetics automatically.

Between the solid solution hcp(Dy) and the β-DyH_2, the PCI measurements were carried out at six different temperatures from 650 °C to 900 °C and between the two hydrides at four different temperature from 250 °C to 325 °C. To do so, a Dy ingot ~0.5 g, which allows determining the H/M value from the pressure changes with sufficient accuracy, was loaded into a tungsten foil boat to prevent contact with the stainless steel reactor, following a baking out at 800 °C under vacuum to completely remove residual hydrogen in the specimen. For most metals or alloys, such as Mg, Ti or TiFe, several cycles of activation process of hydrogenation and dehydrogenation is needed before hydrogen absorption measurement due to poor kinetics and compact oxidation layer [30,31]. For RE-H binary system, however, the hydrogen atoms could pass through the loose rare earth oxide layer, and then diffuse into the metallic layer underneath with excellent hydrogen absorption kinetics [32]. So most of the hydrogen absorption curves of high purity RE metal can be measured at the first cycle of hydrogenation without any activation treatment [9,33–37]. The reproducibility of the results can be monitored by measuring the first isotherm more than once.

The hydrogen dosing was manually controlled using a needle valve and elaborately controlled so that the start and end points of each plateau are more accurately determined. The equilibrium hydrogen pressure was measured after the pressure reached constant for at least 0.5 h and the reproducibility was confirmed by measuring the first isotherm more than once. After each measurement, a new dysprosium sample was used in order to avoid possible cumulative oxygen contamination. For RE-H system, desorption isotherms are very hard to measure because of long equilibrium times. So most of the PCI curves of the RE–H binary systems were investigated by hydrogen absorption measurements [9,33–37]. Coupled studies of absorption PCI curves and CALPHAD method was carried out in La-H [38], Nd-H [6], Gd-H [8] and Er-H [9] systems and self-consistent thermodynamic databases of these systems have been obtained. So we only measure absorption isotherms in this work. Since there was a big difference in temperature between the sample reactor and the other part of the apparatus, the hydrogen content was calculated by volumetric (Sieverts) method [39], the most common technique used for hydrogen adsorption measurement. By comparing the result of PCI curves for Gd obtained in our recent study [8] with those reported in the literature [33], the reliability and accuracy of this system was proved.

The Dy-H samples with different H/Dy ratio synthesized at

650 °C or 300 °C are analyzed by X-ray Diffraction (XRD, PA Nalytical X'Pert3 Powder, Cu Kα, 2°/min). After the synthesis, the stainless steel sample reactor can be opened under ambient atmosphere due to the very high stability of the hydride phases in air and the absence of pyrophoricity.

Differential scanning calorimetry (DSC) curves were obtained with a Netzsch DSC 204 HP calorimeter using 5–6 mg as-prepared DyH_3 powders in aluminum pans, in flowing high purity hydrogen gas, at 50 mL min^{-1} and with a heating rate of 2 °C min^{-1} from room temperature to 500 °C. The temperature of phase transition was determined under 1.2 bar and 5.0 bar H_2 respectively.

Finally, the Dy-H binary phase diagram was assessed with the CALPHAD method using the Pandat phase equilibrium calculation software.

4. Results and discussion

4.1. Characterization of Dy-H binary system

4.1.1. Hydrogen induced phase transition

Six Dy-H samples with representative H/Dy ratio are characterized by XRD, as presented in Fig. 1. Both the metallic Dy and pulverous $DyH_{2.97}$ sample have typical hexagonal structure (ICSD 43581 and ICSD 629827) while $DyH_{1.82}$ shows a face-centered cubic structure (ICSD 629826). Even before the emergence of the β phase (H/M = 0.12), H incorporation in metallic Dy is clearly suggested by the shift of the diffraction peaks to the lower angle, indicating lattice expansion. As the metal becomes saturated with hydrogen (H/M = 1.08), additional hydrogen initiates the formation of the dihydride phase. With a hydrogen content of 2.02 H/M, the metal is completely transformed to β phase with a face-centered cubic structure. After hydrogenation to $DyH_{2.79}$, DyH_3 peaks appears alongside the DyH_2 peaks, thus showing the presence of both DyH_2 and DyH_3 in the sample. The lattice contraction leads to right offset of DyH_2 peaks with increasing hydrogen content. This phenomenon is difficult to understand from a topological point of view but was well confirmed by previous X-ray diffractometry [40–42]. Finally, with a hydrogen content of 2.97 H/M, the sample is completely hydrogenated into DyH_3.

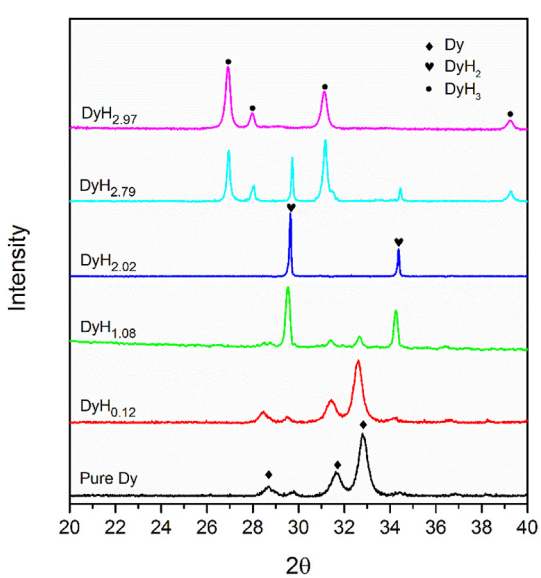

Fig. 1. XRD patterns of phase transformations upon hydrogenation.

4.1.2. Pressure-composition-isotherms

The hydrogenation isotherms, conducted in the hcp(Dy)+β-DyH_2 two phase region, are measured at six different temperatures from 650 to 900 °C, as illustrated in Fig. 2. For Dy at 800 °C, for instance, the plateau corresponding to the Dy/DyH_2 biphasic region between H/Dy from 0.41 to 1.65 occurs at a constant pressure of 1.6 Pa. The width of the plateau gradual shrinks with increasing temperature, indicating a higher solubility of hydrogen in the solid solution hcp(Dy) and a higher fraction of vacant sites in the hypostoichiometric β-DyH_2 at high temperature. The above features are characteristic of a typical hydrogen induced first order phase change behavior in RE-H system.

Fig. 3 presents PCI curves measured between the two hydrides at four different temperature from 250 °C to 325 °C. One well defined plateau between H/Dy from 2.3 to 2.9 is observed for each temperature, which corresponds to the DyH_2/DyH_3 biphasic region. The composition ranges of the second plateaus are in very good agreement with previous results determined by X-ray diffractometry [42].

In our measurement, the hydrogen dosing is elaborately controlled so that the start and end points of each plateau are more accurately determined compared to previous studies, which allows more accurate determination of the phase boundaries.

4.1.3. Thermodynamic properties of phase transformation

The flat plateau in the PCI curves is characteristic of a phase transition. Thermodynamic properties of phase transformation can be determined from the well-known van't Hoff equation for the temperature dependence of the plateau pressures:

$$\ln P_{eq} = \frac{\Delta H_f^0}{RT} + \frac{\Delta S_f^0}{R} \quad (1)$$

where ΔH_f^0 and ΔS_f^0 are the enthalpy and entropy of the hydride formation, respectively. Van't Hoff plots for the temperature dependence of the plateau pressures are shown in Fig. 4 compared with McQuillan's data [27]. The excellent linearity between $\ln P_{eq}$ and 1/T is evidence of the high accuracy of the PCI measurement. From the slope, the values of ΔH_f^0 for hcp(Dy)→β-DyH_2 and β-DyH_2→γ-DyH_3 are 213.1 ± 2.8 kJ/mol H_2 and 79.3 ± 2.2 kJ/mol H_2, respectively. The former result is in very good agreement with the value of 209.4 kJ/mol H_2 in McQuillan's work [27], which was obtained by direct measurement of the van't Hoff plot by temperature variation with a fixed H/M ratio in the reactor.

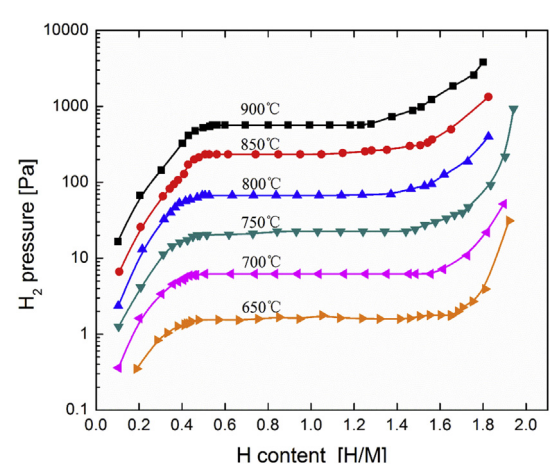

Fig. 2. Pressure-composition absorption isotherms measured from 650 °C to 900 °C in H/Dy = 0.0–2.0 (hcp(Dy) and β-DyH_2).

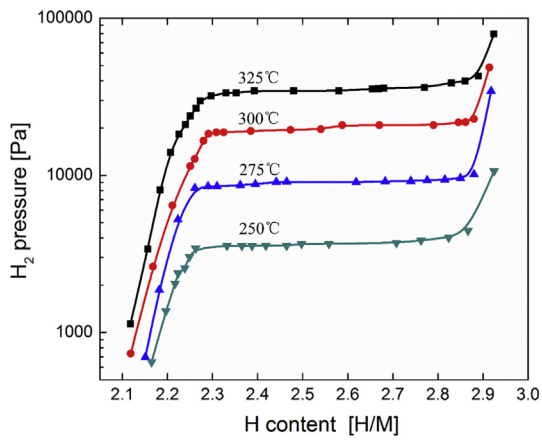

Fig. 3. Pressure-composition absorption isotherms measured from 250 °C to 325 °C in H/Dy = 2.0–3.0 (β-DyH_2 and γ-DyH_3).

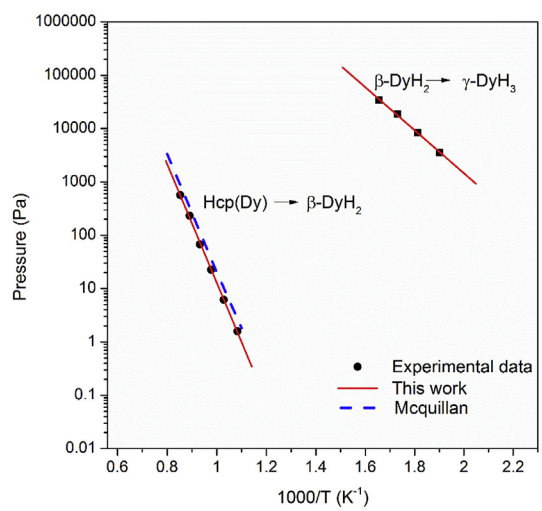

Fig. 4. Van't Hoff plots for the Dy-H system compared to McQuillan's measurements

4.2. Thermodynamic assessment of the binary Dy-H system

With the experimental thermodynamic and phase boundary data in this work, the Dy-H binary phase diagram can be assessed with the CALPHAD method. This extensively used semi-empirical method describes the Gibbs energies of the different phases in Dy-H system with a set of adjustable parameters. These parameters are fitted using the experimental data, and then the phase boundaries and other thermodynamic parameters of interest can be obtained.

4.2.1. Thermodynamic modeling of the Dy-H binary system

The Gibbs energy functions of the pure elements (Dy and H) are taken from the SGTE (Scientific Group Thermo data Europe) unary database [20] and described by an equation of the following form:

$$^{0}G(T) = a + bT + cTln(T) + dT^2 + eT^3 + \frac{f}{T} + \sum_n g_n T^n \quad (2)$$

where a to f and g_n are documented coefficients. No data being available in the SGTE database for the metastable dysprosium as pure fcc. So the lattice stability of Dy in fcc is assessed in this work.

The di-hydrogen H_2 is the only specie considered in the gas phase and the Gibbs free energy was given by:

$$G^{gas} = {}^{ref}G^{gas}_{H_2} + RTln(P_{H_2}/P_0) \quad (3)$$

Below 1381 °C, the crystal structure is hexagonal closest packing (hcp) for metallic Dy while is face centered cubic (fcc) for β-DyH_2. Incorporation of atomic H in Dy metal can be regarded as filling the H into interstitial sites [22,43]. In both fcc and hcp structure, there are one relatively larger yet more weakly absorbing interstitial octahedral (O) site and two more energetically favorable interstitial tetrahedral (T) sites per Dy atom. β-DyH_2 with CaF_2 crystal structure can be hypostoichiometric and hyperstoichiometric. For β-DyH_{2+x}, the tetrahedral sites are completely filled with hydrogen, with the octahedral sites accommodating the remaining x hydrogen [24]. So a three sublattice model is needed for the fcc phase. The HoD_3 structure γ-DyH_3 derives from a hcp structure in which all the tetrahedral sites are occupied by hydrogen and last third of hydrogen is located for 1/3 in triangular sites and for 2/3 in displaced octahedral sites [22,23]. Because this structure is very close to the hcp structure, a three sublattice model is used to simplify the model for the Dy-H solid phases. The model is treated as the formula $(Dy)(H,Va)^O(H,Va)^T_2$ in which the second and third interstitial sublattices represent the octahedral and tetrahedral sites, respectively. The term $(H,Va)^O$ means that the O sites can be either occupied by a H atom (H) or unoccupied (Va), which is similar for the T sublattice. The same thermodynamic model is used for hcp(Dy) and γ-DyH_3 because their crystal structure are similar. If the hydrogen atoms occupied the same interstices with same fraction, the Gibbs free energy of formation should be same. So both the hcp(Dy) and γ-DyH_3 are described as a single hcp phase, consistently with their crystal structures.

No experimental data is available to identify interstitial locations of hydrogen in the bcc phase. In this work, the bcc phase is described by a two sublattice model $(Dy)(H,Va)_u$ with one sublattice fully occupied by Dy and the other one occupied with hydrogen and vacancies. The ratio between sublattices u is gererally based on the number of occupied interstitial sites per dysprosium atom. If the tetrahedral (T) sites were assumed to be occupied in the bcc form, the value of u should be 6. However, the actual value of u for bcc form is generally smaller in order to avoid more model parameters for describing the large excess Gibbs energies and site blocking effect [19,44]. Thus the value of u is assumed to be 1, equal to that used in Sm-H [7] and Gd-H [8]system for the bcc phase.

The description of the Gibbs energy of a phase φ is based on the following equations:

$$G^{\varphi} = {}^{ref}G^{\varphi} + {}^{id}G^{\varphi} + {}^{ex}G^{\varphi} \quad (4)$$

Where $^{ref}G^{\varphi}$ is the contribution of pure components to the Gibbs energy, $^{id}G^{\varphi}$ is the ideal mixing contribution and $^{ex}G^{\varphi}$ is the contribution due to non-ideal interactions between the components, also known as the excess Gibbs energy of mixing.

For φ = hcp and fcc, where two interstitial lattices represent tetrahedral and octahedral sites:

$$^{ref}G^{\varphi} = y'_{Va}y''_{Va}G^{\varphi}_{Dy:Va:Va} + y'_{H}y''_{Va}G^{\varphi}_{Dy:H:Va} + y'_{Va}y''_{H}G^{\varphi}_{Dy:Va:H} + y'_{H}y''_{H}G^{\varphi}_{Dy:H:H} \quad (5)$$

$$^{id}G^{\varphi} = RT\left(\left(y'_{H}lny'_{H} + y'_{Va}lny'_{Va}\right) + 2\left(y''_{H}lny''_{H} + y''_{Va}lny''_{Va}\right)\right) \quad (6)$$

$$^{ex}G^{\varphi} = y'_H y'_{Va} y''_{Va} L^{\varphi}_{Dy:H,Va:Va} + y'_H y'_{Va} y''_H L^{\varphi}_{Dy:H,Va:H} + y'_H y''_H y''_{Va} L^{\varphi}_{Dy:H:H,Va} + y'_{Va} y''_H y''_{Va} L^{\varphi}_{Dy:Va:H,Va} \quad (7)$$

where y'_H and y'_{Va} represents are the site fractions of H and Va in the second sublattice and y'' refers to site fraction in the third. sublattice $G^{\varphi}_{Dy:O:T}$, where O and T could be either H or Va, are the Gibbs energies of the "end-members". L^{φ} is the binary interaction parameters that can be described as linear functions of temperature: $L = A + BT$, in which A and B are coefficients optimized in the binary models.

For φ = bcc, where only one interstitial lattice is present:

$$^{ref}G^{\varphi} = y'_{Va} G^{\varphi}_{Dy:Va} + y'_H G^{\varphi}_{Dy:H} \quad (8)$$

$$^{id}G^{\varphi} = RT\left(y'_H \ln y'_H + y'_{Va} \ln y'_{Va}\right) \quad (9)$$

$$^{ex}G^{\varphi} = y'_H y'_{Va} L^{\varphi}_{Dy:H,Va} \quad (10)$$

where y'_H and y'_{Va} represents the site fraction of the H occupied and vacant second sublattice sites. $G^{\varphi}_{Dy:Va}$ and $G^{\varphi}_{Dy:H}$ are the expressions for the Gibbs energy of the pure bcc Dy and the hypothetical compounds DyH_1 with bcc arrangements of Dy atoms, respectively.

The liquid phase has not been considered due to the absence of any experimental data.

4.2.2. Thermodynamic optimization

The optimization of model parameters and the calculation of phase diagrams were carried out with the Pandat software. Our experimental data of phase boundaries and thermodynamic properties were selected for the evaluation of the thermodynamic model parameters. In addition, experimental information in the literature [21,28,29,42,45] was considered except the PCI curves measured by Beck [25] due to the insufficient accuracy and rough determination of phase boundaries. During an initial guess with the fitting parameters, the entropy of formation for the atomic H is 65 J mol^{-1} K^{-1}. The thermodynamic parameters for fcc and hcp phases are optimized at the first stage. Then, the optimization is continued by taking into account bcc phase. Due to the lack of any experimental data in this phase, for the enthalpy of formation of the end-member DyH (bcc) as a first approximation, the value of the DyH fcc phase has been used, i.e. $\Delta H_{DyH_1}(bcc) = -46654$ J·mol^{-1}. All the parameters were finally evaluated together to give the best description of the system.

According to the thermodynamic description and the optimized parameters listed in Table 1, the calculated Dy-H binary phase diagram at 1 atm with experimental data is presented in Fig. 5(a). Fig. 5(b) shows the enlarged β-DyH_2+γ-DyH_3 mixed phase regions. The experimental data determined by Beck, which are not chosen for optimization, are also used for comparison. It can be seen that the calculated phase diagram is in excellent consistency with our experimental data. It also fairly well reproduces the experimental phase boundaries from most of the previous works. The calculated phase boundaries deviate from Beck's data at high temperature, particularly for the metal-rich phase boundary of non-stoichiometric dysprosium dihydride, as the initial and ending points in the PCI plateaus in Beck's data are only roughly estimated. Nevertheless, the general agreement with the published data is fairly good. Without considering the liquid phase, two calculated invariant reactions in this evaluation are listed in Table 2.

In order to check the quality of the modeling, PCT curves are simulated using the obtained parameters in Table 1. As shown in Fig. 6, the agreement between the calculated results and the experimental data is fairly good in the phase regions from pure dysprosium up to dysprosium trihydride.

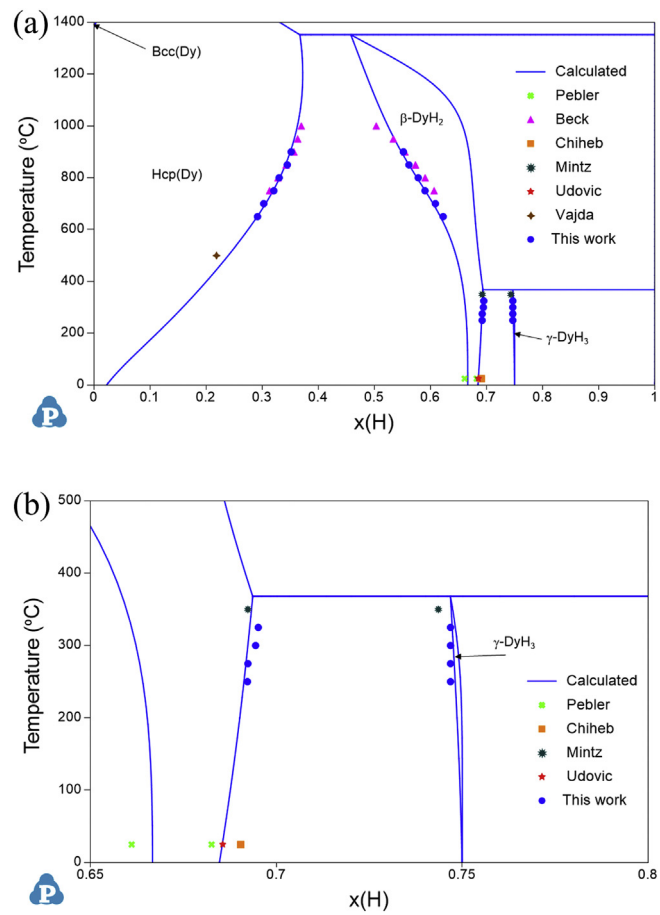

Fig. 5. (a) The calculated Dy-H binary phase diagram at 101325 Pa compared with some experimental measurements and (b) enlarged β-DyH_2 + γ-DyH_3 mixed phase regions.

Table 1
The optimized parameters of the binary Dy-H system.

Phase	Parameters
hcp	$G^{Hcp}_{Dy:Va:Va} = G^{SER}_{Dy}$ $G^{Hcp}_{Dy:H:Va} = G^{SER}_{Dy} + G^{SER}_{H} - 65831.08 + 65 \times T$ $G^{Hcp}_{Dy:Va:H} = G^{SER}_{Dy} + 2 \times G^{SER}_{H} - 169184 + 130 \times T$
	$G^{Hcp}_{Dy:H:H} = G^{SER}_{Dy} + 3 \times G^{SER}_{H} - 252710.72 + 195.99 \times T$ $^0L_{Dy:*:H,Va} = -23400.769 + 5.58 \times T$ $^0L_{Dy:H,Va:Va} = -21425.17$
fcc	$G^{Fcc}_{Dy:Va:Va} = G^{SER}_{Dy} + 5562 - 0.83 \times T$ $G^{Fcc}_{Dy:H:Va} = G^{SER}_{Dy} + G^{SER}_{H} - 49871 + 65 \times T$
	$G^{Fcc}_{Dy:Va:H} = G^{SER}_{Dy} + 2 \times G^{SER}_{H} - 213075.33 + 138.5 \times T$ $G^{Fcc}_{Dy:H:H} = G^{SER}_{Dy} + 3 \times G^{SER}_{H} - 230708.11 + 195 \times T$ $^0L_{Dy:*:H,Va} = 43836.757 - 19.891767 \times T$
	$^0L_{Dy:H,Va:*} = -21425.169$
bcc	$G^{Bcc}_{Dy:Va} = G^{Bcc}_{Dy}$ $G^{Bcc}_{Dy:H} = G^{Bcc}_{Dy} + G^{SER}_{H} - 46654 + 65 \times T$

Table 2
Invariant reactions in the Dy-H system at 101325 Pa hydrogen pressure.

Reaction	T (°C)	Composition (mol% H)		
β-DyH_2+Gas(H_2)→γ-DyH_3	367.9	0.694	1.000	0.747
Hcp(Dy) +Gas(H_2)→β-DyH_2	1352.6	0.367	1.000	0.458

The dehydrogenation of DyH_3 is further studied by DSC in hydrogen gas flow (50 mL min^{-1}, 1.2 bar or 5.0 bar) and the result is shown in Fig. 7. Under hydrogen pressure of 1.2 bar, a clearly endothermic peak corresponding to the transition $DyH_3 \rightarrow DyH_2 + H_2(gas)$ is observed at about 372.1 °C, very close to the 371.2 °C gained by CALPHAD method. While under 5.0 bar (Fig. 7(b)), the endothermic peak shift toward high temperature (442.7 °C) due to the increasing stability of DyH_3, in good agreement with the calculated decomposition temperature (444.9 °C).

As shown in Fig. 8(a), the proportions of all the three solid phases at 25 °C are calculated using the obtained parameters in Table 1. The calculated values are satisfactorily consistent with the measured data in Ref. [21]. The calculated hydrogen occupancy in the different interstitial sites is presented in Fig. 8(b) compared with the experimental data. When H/M < 2, only the tetrahedral sites are occupied by H atoms, while for H/M > 2 the octahedral sites accommodate the additional hydrogen. This result is consistent with the characteristic occupancy of the heavy rare-earth metals at room temperature [21,41,46]. The calculated result is in good agreement with the experimental data obtained by Udovic [21].

In view of their chemical properties the lanthanide metals are divided into light (La, Ce, Pr, Nd and Sm) and heavy (Gd, Tb, Dy, Ho, Er, Tm, Yb and Lu). Three heavy rare earth-hydrogen binary systems (Gd-H [8], Dy-H and Er–H [9]) have been optimized thermodynamically by the CALPHAD technique. The Dy-H phase diagram calculated in the present work is similar to that of Gd-H and Er-H systems reported in the literature. These binary systems show high solubility of hydrogen in the RE metal. The solubilities of hydrogen in these heavy rare earth metals increase rapidly with temperature. The β-REH_2 phase presents a large extension at low hydrogen content and at high temperature. And the γ-REH_3 phase has a small composition domain. In the meantime, of course, there are small differences between the position of phase boundaries and the dissociation pressure of β-REH_2 and γ-REH_3. When other important systems (Tb-H, Ho-H, Tm-H and Yb-H) are calculated, variation of the heavy rare earth-hydrogen phase diagrams with the atomic number can be derived.

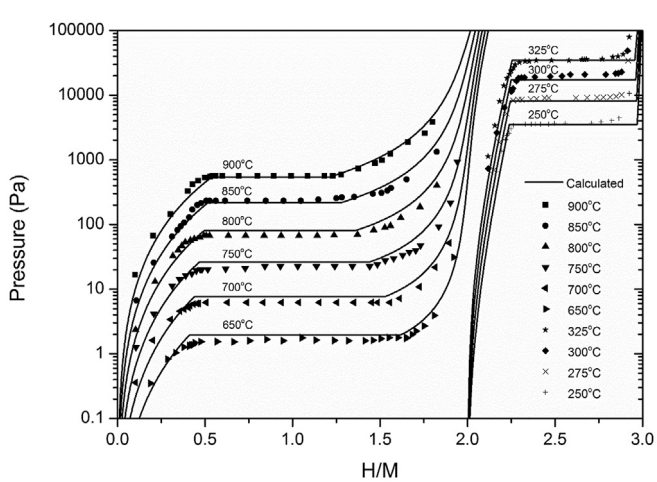

Fig. 6. Calculated PCT curves compared with experimental measurements.

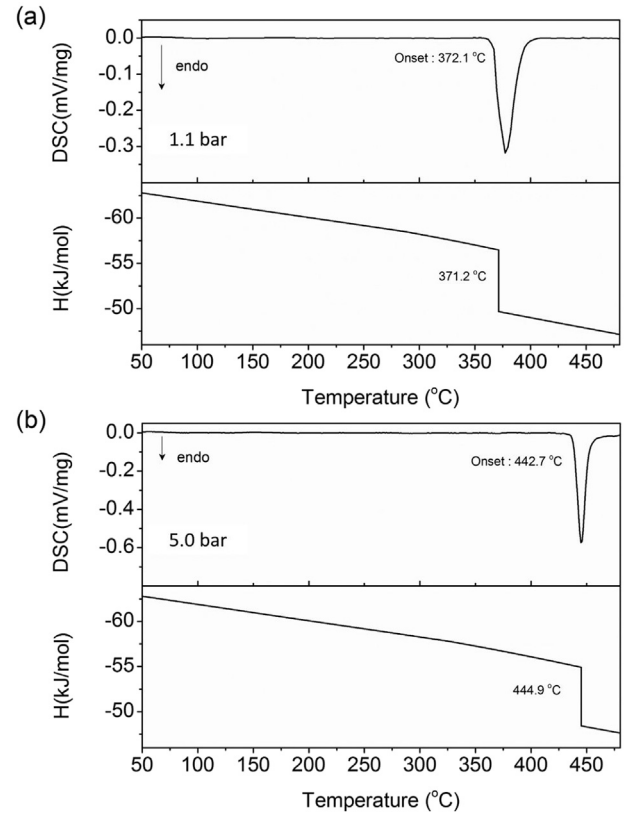

Fig. 7. DSC curve and calculated enthalpy of the reaction $DyH_3 = DyH_2 + 1/2H_2$ at (a) 1.1 bar and (b) 5.0 bar hydrogen pressure.

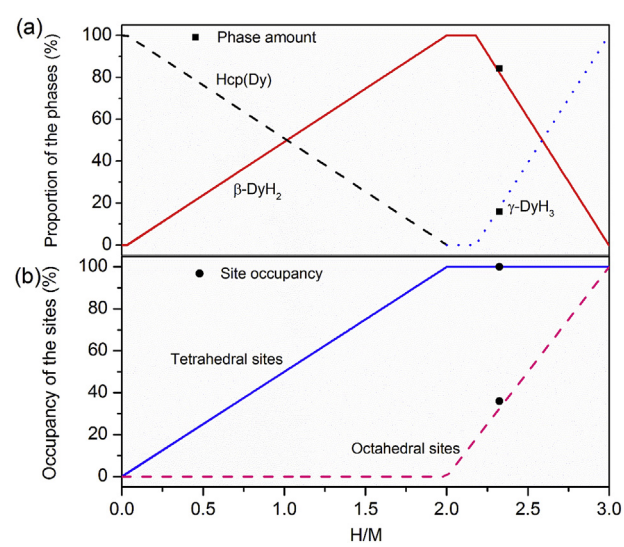

Fig. 8. (a) Evolution of the proportions of the three solid phases as function of the H/M composition at 25 °C, compared with experimental data [21] and (b) Evolution of the occupancy of the interstitial sites as a function of the H/M composition at 25 °C, compared with experimental data [21].

The Gd_xDy_{1-x}-hydrogen system is worth to investigate in order to compare the effect of alloying on the hydrogen solubility and hydride formation in the lanthanide metals alloy. Up to now, no multicomponent lanthanide alloy-hydrogen systems have been described by the CALPHAD technique. And problems of

compatibility have been noticed [10]. The compatible models between the three binaries should be used if systems are to be combined. We have choose the same model for both Gd-H and Dy-H system, which will make the operation convenient to combine simple systems in order to obtain predictions of phase equilibrium in multicomponent systems.

5. Conclusions

A combined study by PCI measurement and phase diagram calculation is carried out to provide more quantitative understanding on the thermodynamics of the dysprosium-hydrogen binary system. The phase boundaries are determined by PCI measurement with high accuracy. Between the Dy and the dihydride β-DyH_2 the measurements have been made from 650 °C to 900 °C in the 10^{-1}-10^5 Pa pressure range. The composition ranges of the plateaus corresponding to the Dy/β-DyH_2 biphasic region are H/Dy = 0.41–1.65 at 650 °C, much broader than the range H/Dy = 0.54–1.23 at 900 °C. The phase boundaries obtained are in agreement with previous experimental work but with improved accuracy. β-DyH_2/γ-DyH_3 two-phase plateau region is observed in the composition ranges of H/M around 2.3–2.9. A calculated Dy-H binary phase diagram is obtained by the CALPHAD technique based on our experimental results and critically evaluated experimental information available in the literature. The calculated phase equilibria and thermodynamic quantities agree rather well with experimental data.

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